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NC - 14 (L1)

Frontend Assignment

Ans 1 CSS Box Model

The CSS Box model is a fundamental concept that describes how every HTML element is displayed as a rectangular box on a web page.

It consists of four parts:

1. Content - The actual text, image or data inside the element.
2. Padding - Space between the content and the border.
3. Border - A line surrounding the content and padding.
4. Margin - Space outside the border that separates elements from each other.

The total size of an element is calculated by combining content, padding, border, and margin.

Ans 2 media query in CSS

It is used to make a website responsive. allows different styles to be applied based on screen size, device type or resolution.

Example:

```
body {  
  background-color: white;  
}
```

```
@media (max-width: 600px) {
```

```
  body {
```

```
    background-color: lightblue;  
  }
```

In this example, the background color changes when screen width becomes 600px or smaller.

Ans 3 Important Features of Bootstrap

Bootstrap is a popular front-end framework used for creating responsive websites quickly.

Features:

- Responsive grid system
- Predefined CSS classes
- Ready-made components such as buttons, cards, forms, and navigation bars
- Built-in JavaScript components
- Mobile-first design approach
- Easy customization
- Faster web development

Bootstrap reduces development time and ensures consistency across web pages.

Ans 4 Bootstrap Grid System

Bootstrap uses a 12-column grid system to create responsive layouts.

Example:

```
<div class = "container">  
  <div class = "row">  
    <div class = "col-4"> column 1 </div>  
    <div class = "col-4"> column 2 </div>  
    <div class = "col-4"> column 3 </div>  
  </div>  
</div>
```

Explanation

- container creates main layout area.
- row creates horizontal row
- col-4 occupies 4 out of 12 columns.
- col-4 occupies 4 out of 12 columns. together occupy all 12 columns.

Ans 5 CSS selectors

CSS selectors are used to target HTML elements and apply style

(a) Element selector

```
p {
```

```
  color: red;
```

```
}
```

styles all paragraph elements.

(b) class selector

```
.box {
```

```
  color: blue;
```

```
}
```

styles all elements having class "box".

(c) ID selector

```
#heading {
```

```
  color: green;
```

```
}
```

styles the element with id "heading".

(d) universal selector

```
* {
```

```
  margin: 0;
```

```
}
```

applies style to all elements.

(e) Group selector

```
h1, p {
```

```
  color: black;
```

```
}
```

applies same style to multiple elements

Ans 6 CSS Background Properties

Background properties are used to control the appearance of an element's background.

Important properties are:

- background-color
- background-image
- background-repeat
- background-position
- background-size
- background-attachment
- background

These properties help improve the visual appearance of web pages.

Ans 7 overlapping Elements in CSS

Elements can be overlapped using the position and z-index properties.

Example:

• box 1 $\{$
position: absolute;
left: 50px;
top: 50px;
z-index: 1;
}

• box 2 $\{$
position: absolute;
left: 80px;
top: 80px;
z-index: 2;
}

The element with the higher z-index value appears above the element with the lower value.

Ans 8 CSS Position Property

The position property specifies how an element is placed on a web page.

Types of Position:

1. Static

Default position of an element.

`position: static;`

2. Relative

Moves relative to its original position.

`position: relative;`

`top: 20px;`

3. Absolute

Positioned relative to the nearest positioned ancestor.

`position: absolute;`

`top: 20px;`

4. Fixed

Remains fixed even when the page is scrolled.

`position: fixed;`

`bottom: 10px;`

5. Sticky

Acts like relative until a certain scroll position is reached.

`position: sticky;`

`top: 0;`

Ans 9 ECMA Script

ECMA Script is the standard specification on which JavaScript is based. It defines the syntax, features, and behaviour of the JavaScript language.

ECMA Script includes:

- Variables
- Data Types
- Functions
- Objects
- Classes
- Loops
- Modern JavaScript Features

Example:

ES6 (ECMAScript 2015) introduced several important features such as:

- let
- const
- Arrow functions
- Classes
- Template literals

These features make JavaScript more powerful and easier to use.

Ans 10 Difference between var, let, and const

Feature	var	let	const
Scope	Function Scope	Block Scope	Block Scope
Redeclaration	Allowed	Not Allowed	Not Allowed
Reassignment	Allowed	Allowed	Not Allowed
Hoisting	Yes	Yes	Yes
Temporal Read zone	No	Yes	Yes

Temporal Read zone (TDZ)

is the time period between entering a block and declaring a let or const variable. During this period, variable exists but cannot be accessed.

Example

```
console.log(x);  
let x = 10;
```

This code produces an error because x is accessed before its declaration.